

Aspect	Traditional Systems Engineering	COTS-Based Considerations
System Architecture and Design	Focus is on optimizing the SoI	Focus is on optimizing the set of COTS and custom components that make up the SoI
Integration, Verification, and Validation	Done with known (internal) system element owners	Can typically get an early version of the system up and operating dramatically sooner than with a “make” system, criteria more difficult to establish since COTS performance must satisfy the market requirements while balancing the needs of the system, execution and defect resolution more difficult due to external COTS element owners
Technical Management	Well understood set of processes	More challenging decision environment, additional risks are present for COTS, and potentially increased configuration and information management (CM and IM) activities
Agreements	Acquisition agreements are primarily	Acquisitions also include COTS items and must consider other aspects such

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	outsourced development efforts	as licensing, additional COTS vendor support, and obsolescence management
Quality Characteristics	Known with internal team support and data, but the full picture may not be obtainable until after system deployment	For proven COTS, can be determined up front, may have to rely on COTS vendors for some data, consideration of life cycle cost (LCC) is critical for COTS

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4.3.4 Software-Intensive Systems

ISO/IEC/IEEE 42010 (2022) defines a software-intensive system as:

Any system where software contributes essential influences to the design, construction, deployment, and evolution of the system as a whole.

Software, like physical entities, is encapsulated in system elements. Software elements often contribute to system functionality, behavior, quality characteristics, interfaces, and observable performance indicators for software-intensive systems. Encapsulated software is sometimes custom-built and is sometimes obtained by using software elements from libraries. Reused software elements may be modified by tailoring them for

intended use. Also, software elements may be licensed from software vendors. In most cases, licensed software packages cannot be modified. Software engineers sometimes encapsulate licensed software packages in software shells or wrappers that provide the interfaces needed to integrate the needed capabilities of the software into a system while masking unwanted capabilities.

Software is widely incorporated in systems and provides “essential influences” because software is a malleable entity composed of textual and iconic symbols that in many cases, but not always, can be constructed, modified, or procured sooner and at less expense than fabricating, modifying, or procuring physical elements that have equivalent capabilities. In some cases, but not always, software elements can provide capabilities that would be difficult to realize in hardware.

But in some cases, a physical element of equivalent capability may be preferred to a software element because software may not provide certifiable safety, security, or performance at the necessary levels of assurance. Making tradeoffs between physical elements and software elements is an essential aspect of developing and modifying software-intensive systems. Trade studies are best conducted when SE practitioners consult physical engineers and software engineers who are working collaboratively.

Interfaces provided by software in a software-intensive system can provide passive pass-through connections or, if desired, software included in an interface can actively coordinate interactions among the connected elements. Software interfaces may be internal and/or external to a system. Internal software interfaces can provide connections that coordinate interactions among physical elements, among software elements, and between physical elements and software elements. External software interfaces can provide passive and active connections to entities in the physical environment, including humans and other sentient entities that are software-enabled, plus connections to the external interfaces of other physical-only, software-only, and software-intensive systems. A “software-

only” system is a software application implemented on a stable computing platform. The computing platform includes hardware and software that support the software application. The platform is often implemented using commodity items.

Sometimes the essential influences of software are observable in a system’s external interfaces, as in human-user interface displays, and sometimes not, as in the interfaces for direct interaction with an external system. In the latter case, the software is said to be embedded in the SoI because performance of the interface is not directly observable by a human, although the effects of interface performance may be observable. Software is also used to provide interfaces among the systems that constitute an SoS. Software can contribute to functionality, behavior, quality attributes, external interfaces, and performance indicators for a composite SoS.

Developing and modifying software-intensive systems presents challenges for SE practitioners when partitioning system requirements and elements of the architecture, allocating performance parameters to physical elements and to software elements, establishing and controlling physical-software interfaces, and facilitating integration of physical elements with software elements (Fairley, 2019). These challenges sometimes arise because the culture, terminology, processes, and practices of software engineering are unfamiliar to SE practitioners and conversely, the various aspects of SE may be unfamiliar to software engineers. Techniques for improving communication between SE practitioners and software engineers are presented in [Section 5.3.1](#) and in Fairley (2019).

4.3.5 Cyber-Physical Systems (CPS)

CPS are the integration of physical and cyber (software) processes in which the software monitors and controls the physical processes and is, in turn, affected by them. CPS are enabled by sensors and feedback loops and their provenance has increased because of significant advances in sensor technology and affordability. [Figure 4.6](#) illustrates the behavior of a CPS: sensors may be deployed in the system hardware and/or its environment.

Sensor data is used by software algorithms to control the hardware in responses to changes in the environment and/or in the hardware itself. The algorithms control the dynamic behavior of the CPS to achieve one or more goals, which could include homeostasis (maintaining equilibrium). An example might be an automobile with sensors to detect obstacles (in the environment) and take evasive action if an obstacle is detected ahead (the actuators would be steering or braking). Other sensors may be deployed to monitor the health of the vehicle, for instance, wear of the brake pads may trigger an action to modify the driving or notify the owner of the need for maintenance. Digital twins are also an important concept in this regard that refers to a digital surrogate that is a dynamic physic-based description of physical assets (physical twin), processes, people, places, systems, and devices that can be used for various purposes. The digital representation provides both the elements and the dynamics of how an Internet of Things (IoT) device operates and lives throughout its life cycle (see [Section 4.3.7](#)).

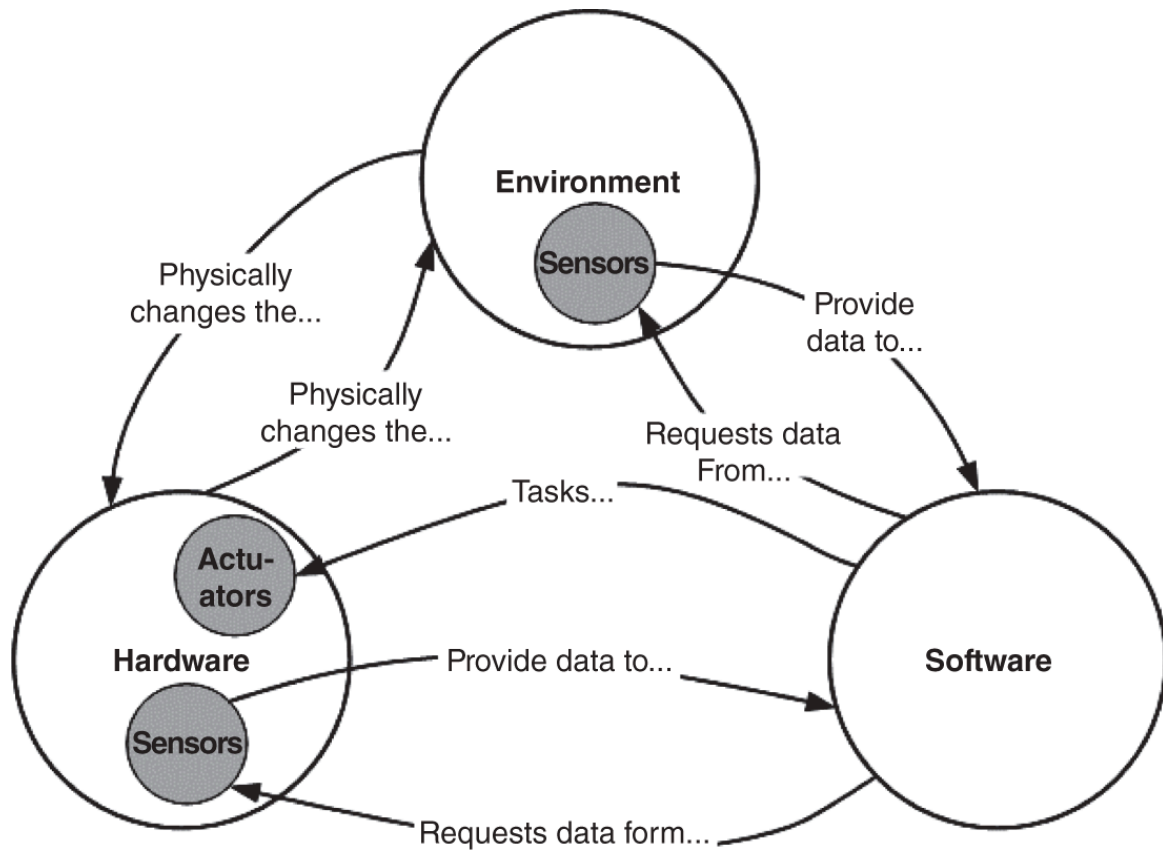


FIGURE 4.6 Schematic diagram of the operation of a cyber-physical System. IncoSE SEH original figure created by Henshaw. Usage per the IncoSE notices page. All other rights reserved.

The CPS concept is closely aligned to Industry 4.0, an initiative to revolutionize industry through so-called smart systems (Kagermann, 2013). CPS always include both software and hardware and are almost always networked, in which case they are Cyber-Physical Systems of Systems (CPSoS). If the individual CPS are networked using Internet Protocols, then they form part of an IoT, but they may interact through other mechanisms (e.g., mechanical, electromagnetic, thermal). The relationships between the concepts of SoS, CPS, and IoT are illustrated in [Figure 4.7](#). The “Things” in IoT are constituent systems and they are always networked, and therefore, always an SoS.

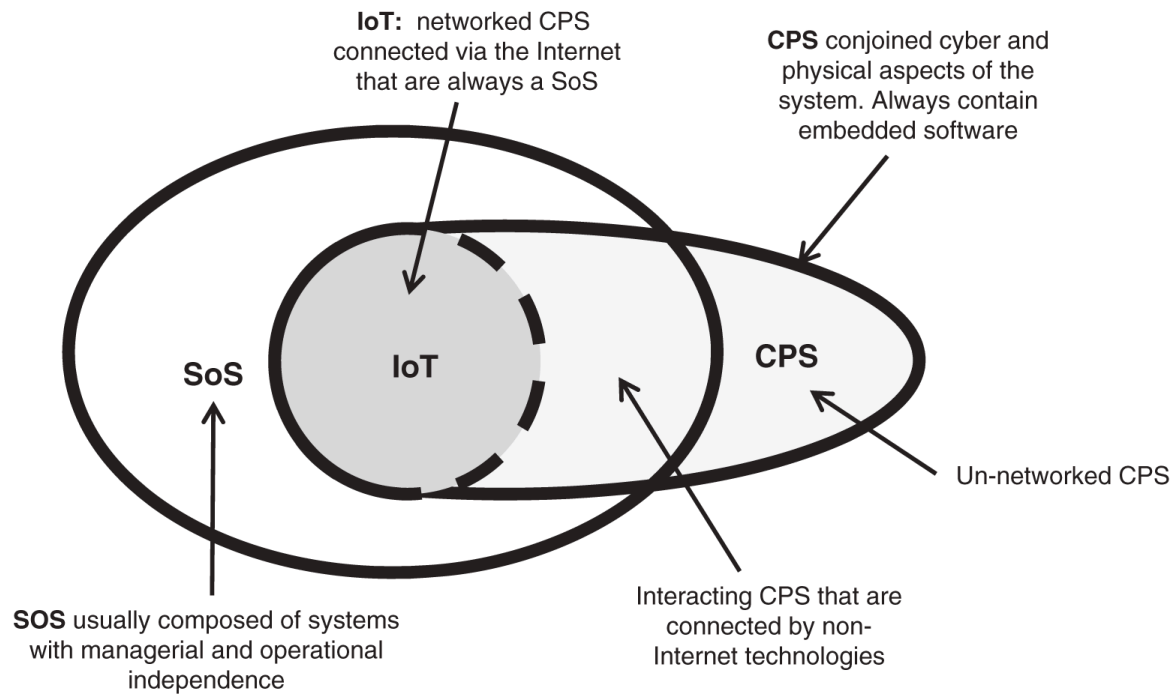


FIGURE 4.7 The relationship between cyber-physical Systems (cpS), Systems of Systems (SoSs), and an Internet of Things (IoT). From Henshaw (2016). Used with permission. All other rights reserved.

CPS are a feature of almost every industry and many aspects of society. They provide automated, and even autonomous, control of technologies ranging from robotic manufacture to SMART cities and from automated insulin delivery to control of critical infrastructure. They may form a contribution to business models that include physical systems and can even underpin a business concept by providing resilience or cost savings.

There are many implications of CPS for engineers. Two of the most significant are complexity and ethics. The increased level of complexity is due to both extensive networks and the problem of modeling the combination of physical dynamics with computational processes. Lee (2015) has pointed out that such models are nearly always nondeterministic due to the lack of temporal semantics in the cyber and physical modeling programs. This has significant implications for modeling and verification within the SE processes. The lack of determinism, together with questions concerning the transfer of decision making from humans to machines have ethical challenges that engineers must face in terms of definition and realization of CPS (European Parliament Research Service, 2016).

The scope and dimensions of CPS is well-illustrated by the CPS Concept Map (Asare, et al., 2012), which essentially defines CPS as feedback systems that are applicable across a wide range of applications such as infrastructure, healthcare, manufacturing, military and many more. One might consider them relevant to any application in which the system is required to be dynamic and the control thereof is managed by software. The concept map also highlights the explicit need for security and safety considerations in design as well as the challenges of verification and validation in large complex systems. It points toward the need for improved modeling and, indeed, progress has been made in the use of MBSE for CPS development through a framework to implement suitable tool chains (Lu, 2019).

Given that software forms an integral part of many systems and devices, it can reasonably be stated that SE practitioners are very often concerned with CPS and usually CPSoS.

4.3.6 Systems of Systems (SoS)

ISO/IEEE/IEC 21839 (2019) defines a System of Systems (SoS) as:

A set of systems or system elements that interact to provide a unique capability that none of the constituent systems can accomplish on its own.

Constituent systems can be part of one or more SoS. Each constituent system is a useful system by itself, having its own development, management goals, and resources, but interacts within the SoS to provide the unique capability of the SoS (ISO/IEEE/IEC 21839, 2019).

The following characteristics can be useful when deciding if a particular SoI can better be understood as an SoS (Maier, 1998):

- Operational independence of constituent systems;
- Managerial independence of constituent systems;
- Geographical distribution;

- Emergent behavior;
- Evolutionary development processes.

Of these, operational independence and managerial independence are the two principal distinguishing characteristics for applying the term SoS.

Figure 4.8 illustrates the concept of an SoS. The air transport system is an SoS comprising multiple aircraft, airports, air traffic control systems, and ticketing systems, which along with other systems such as security and financial systems facilitate passenger transportation. There are equivalent ground and maritime transportation SoS that are all in turn part of the overall transport system (an SoS in the terms of this description).

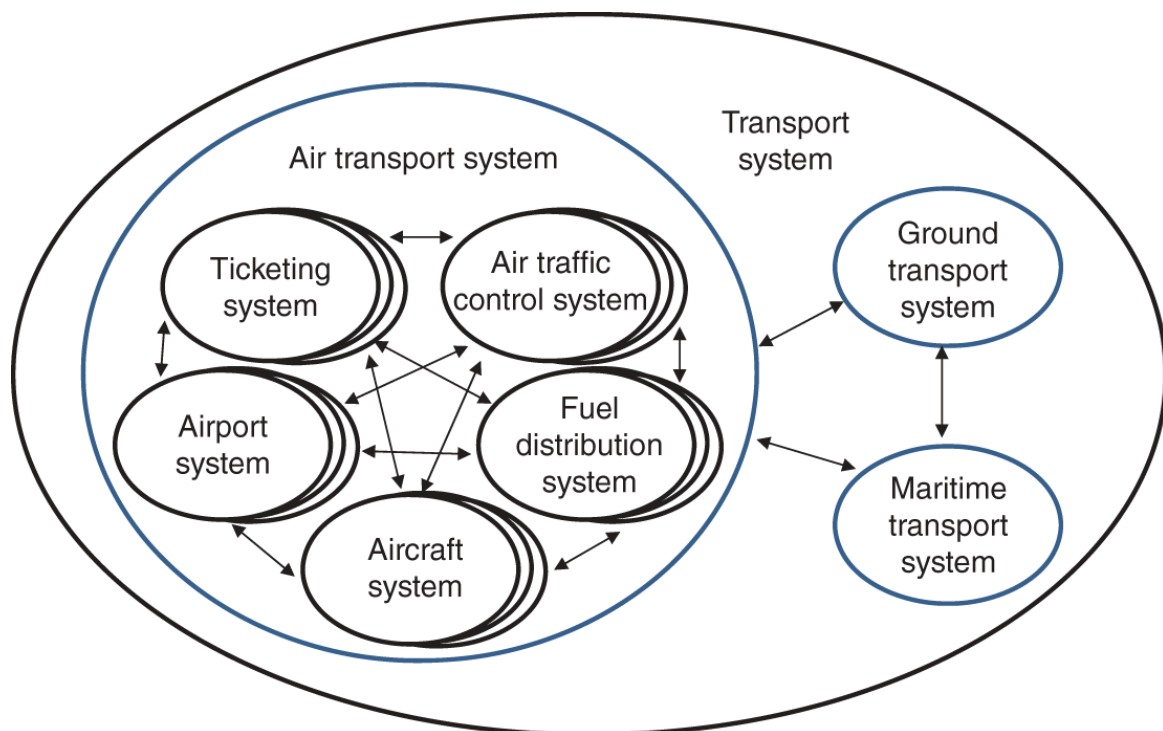


FIGURE 4.8 Example of the systems and systems of systems within a transport system of systems. From ISO/IEC/IEEE 21841 (2019). Used with permission. All other rights reserved.

There are three international standards that provide useful guidance on SoS:

- ISO/IEC/IEEE 21839 (2019)—“System of systems (SoS) considerations in life cycle stages of a system” focuses on the SE of an individual

constituent system and identifies considerations to be addressed as the engineering of the systems progresses from concept through retirement.

- ISO/IEC/IEEE 21840 (2019)—“Guidelines for the utilization of ISO/IEC/IEEE 15288 in the context of system of systems (SoS)” provides guidance on the application of the processes to the special case of SoS, including considerations for how constituent systems relate within the SoS.
- ISO/IEC/IEEE 21841 (2019)—“Taxonomy of systems of systems” lays out a taxonomy of SoS types based on authority relationships as (shown in [Table 4.3](#))

TABLE 4.3 SoS types

Directed	The SoS is created and managed to fulfill specific purposes and the constituent systems are subordinated to the SoS. The constituent systems maintain an ability to operate independently; however, their normal operational mode is subordinated to the central managed purpose.
Acknowledged	The SoS has recognized objectives, a designated manager, and resources for the SoS; however, the constituent systems retain their independent ownership, objectives, funding, and development and sustainment approaches. Changes in the systems are based on cooperative agreements between the SoS and the constituent systems.
Collaborative	The constituent systems interact more or less voluntarily to fulfill agreed upon central purposes. The central players collectively decide how to provide or deny service, thereby providing some means of enforcing and maintaining standards.
Virtual	The SoS lacks a central management authority and a centrally agreed upon purpose for the SoS. Large-scale behavior emerges-and may be desirable-but this type of SoS must rely on relatively invisible mechanisms to maintain it.

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ISO/IEC/IEEE 21840 (2019) guidance on application of ISO/IEC/IEEE 15288 (2023) life cycles processes (see [Section 2.3](#)) is based on the differences

between systems and SoS and the impact on the SE processes as shown in [Table 4.4.](#)

TABLE 4.4 Impact of SoS considerations on the SE processes

SE Process	Implementation as Applied to SoS
Agreement Processes	Because there is often no top level SoS authority, effective agreements among the systems in the SoS are key to successful SoSE.
Organizational Project Enabling Processes	SoSE develops and maintains those processes which are critical for the SoS within the constraints of the system level processes.
Technical Management Processes	SoSE implements Technical Management Processes applied to the particular considerations of SoS engineering - planning, analyzing, organizing, and integrating the capabilities of a mix of existing and new systems into a system of systems capability while systems continue to be responsible for technical management of their systems.
Technical Processes	SoSE Technical Processes define the cross-cutting SoS capability, through SoS level business or mission analysis and stakeholder needs and requirements definition. SoS architecture and design frame the planning, organization, and integration of the constituent systems, constrained by system architectures. Development, integration, verification, transition, and validation are implemented by the systems. with SoSE monitoring and review. SoSE integration, verification, transition and validation applies when constituent systems are integrated into the SoS and performance is verified and validated.

From ISO/IEC/IEEE 21840 (2019) adapted from SEBOK. Used with permission. All other rights reserved.

Dahmann (2014) identified the following challenges that influence the engineering of an SoS:

- *SoS authorities*—In an SoS, each constituent system has its own local “owner” with its stakeholders, users, business processes, and development approach. As a result, the type of organizational structure assumed for most traditional SE under a single authority responsible for the entire system is absent from most SoS. In an SoS, SE relies on crosscutting analysis and on composition and integration of constituent systems, which in turn depend on an agreed common purpose and motivation for these systems to work together toward collective objectives that may or may not coincide with those of the individual constituent systems.
- *Leadership*—Recognizing that the lack of common authorities and funding poses challenges for SoS, a related issue is the challenge of leadership in the multiple organizational environments of an SoS. This question of leadership is experienced where a lack of structured control normally present in SE requires alternatives to provide coherence and direction, such as influence and incentives.
- *Constituent systems*—SoS are typically composed, at least in part, of in-service systems, which were often developed for other purposes and are now being leveraged to meet a new or different application with new objectives. This is the basis for a major issue facing the application of SE to SoS, that is, how to technically address issues that arise from the fact that the systems identified for the SoS may be limited in the degree to which they can support the SoS. These limitations may affect initial efforts at incorporating a system into an SoS, and systems’ commitments to other users may mean that they may not be compatible with the SoS over time. Further, because the systems were developed and operate in

different situations, there is a risk that there could be a mismatch in understanding the services or data provided by one system to the SoS if the particular system's context differs from that of the SoS.

- *Capabilities and requirements*—Traditionally (and ideally), the system engineering process begins with a clear, complete set of initial user requirements and provides a disciplined approach to develop and evolve a system to meet these and emerging requirements. Typically, SoS are composed of multiple independent systems with their own requirements, working toward broader capability objectives. In the best case, the SoS capability needs are met by the constituent systems as they meet their own local requirements. However, in many cases, the SoS needs may not be consistent with the requirements for the constituent systems. In these cases, SE of an SoS needs to identify alternative approaches to meeting those needs either through changes to the constituent systems or through the addition of other systems to the SoS. In effect, this is asking the systems to take on new requirements with the SoS acting as the “user.”
- *Autonomy, interdependence, and emergence*—The independence of constituent systems in an SoS is the source of a number of technical issues when applying SE to an SoS. The fact that a constituent system may continue to change independently of the SoS, along with interdependencies between that constituent system and other constituent systems, adds to the complexity of the SoS and further challenges SE at the SoS level. These dynamics can lead to unanticipated effects at the SoS level leading to unexpected or unpredictable behavior in an SoS even if the behavior of the constituent systems is well understood.
- *Testing*—The fact that SoS are typically composed of constituent systems that are independent of the SoS poses challenges in conducting end-to-end SoS testing, as is typically done with systems. First, unless there is a clear understanding of the SoS-level expectations and measures of those expectations, it can be very difficult to assess the level of performance as the basis for determining areas that need attention or to ensure users of the capabilities and limitations of the SoS. Even when there is a clear understanding of SoS objectives and metrics, testing in a traditional

sense can be difficult. Depending on the SoS context, there may not be funding or authority for SoS testing. Often, the development cycles of the constituent systems are tied to the needs of their owners and original ongoing user base. With multiple constituent systems subject to asynchronous development cycles, finding ways to conduct traditional end-to-end testing across the SoS can be difficult if not impossible. In addition, many SoS are large and diverse, making traditional full end-to-end testing with every change in a constituent system prohibitively costly. Often, the only way to get a good measure of SoS performance is from data collected from actual operations or through estimates based on modeling, simulation, and analysis. Nonetheless, the SoS SE team needs to enable continuity of operation and performance of the SoS despite these challenges.

- *SoS principles*—SoS is an area where there has been limited attention given to ways to extend systems thinking to the issues particular to SoS. The community is beginning to identify and articulate the crosscutting principles that apply to SoS in general and to develop working examples of the application of these principles. There is a major learning curve for the average SE practitioner moving to an SoS environment and a problem with SoS knowledge transfer within or across organizations.

Beyond these general SE challenges, in today's environment, SoS pose particular issues from a security perspective. This is because constituent system interface relationships are rearranged and augmented asynchronously and often involve COTS elements from a wide variety of sources. Security vulnerabilities may arise as emergent phenomena from the overall SoS configuration even when individual constituent systems are sufficiently secure in isolation.

The SoS challenges cited in this section require SE approaches that combine both the systematic and procedural aspects described in this handbook with holistic, nonlinear, iterative methods. There is a growing set of approaches to applying SE to SoS (Cook and Unewisse, 2019). These include SoS life cycle engineering approaches such as the SoS Wave Model (Dahmann, et al., 2011) and the Designing for Adaptability and evolution in

System of Systems Engineering (DANSE). These approaches address both functionality of constituents to create coherent aggregate SoS capability (Axelsson, 2020) as well as management of interfaces among constituents (Hoehne, 2020).

4.3.7 Internet of Things (IoT)/Big Data-Driven Systems

SE is based on engineering requirements, engineering calculations, testing, modeling, and simulations—and all are based on data or data generation. SE practitioners often make decisions based on intuition, previous experience, or qualitative assessments. The 4th Industrial Revolution, with its proliferation of sensors of various types and big data analytics, creates an opportunity for SE, as a discipline, and for SE practitioners, as professionals and decisions makers, to be more data-driven. The following recommendations apply to modern SE tasks and decisions:

- Bring as much diverse data and as many diverse viewpoints to maximize the generation of information quality.
- Use data to develop a deeper understanding of the business context and the problem at hand.
- Develop an appreciation for the impact of variation, both in data and in the overall business.
- Deal with uncertainty, which means that SE also recognizes mistakes.
- Recognize the importance of high-quality data and invest in trusted sources and in making improvements.
- Conduct good experiments and research to supplement existing data and address new questions.
- Recognize the criteria used to make decisions and adapt under varying circumstances.
- Realize that making a decision is only the first step; SE practitioners must keep an open mind and revise decisions if new data suggests a better course of action.
- Work to bring new data and new data technologies into the organization.

- Learn from mistakes and help others to do so, by applying lessons-learned processes.
- As SE practitioners, strive to be a role model when it comes to data, working with leaders, peers, and subordinates to help them become data driven.

There are three general goals in analyzing data:

1. *Prediction*: To predict the response to future values of the input variables.
2. *Estimation*: To infer how response variables are associated with input variables.
3. *Explanation*: To understand the relative contribution of input variables to response values.

Predictive modeling is the process of applying models and algorithms to data for the purpose of predicting new observations. In contrast, explanatory models aim to explain the causality and relationship between the independent variables and the dependent variables. Classical statistics focuses on modeling the stochastic system generating the data. Statistical learning, or computer age statistics, builds on big data and the modeling of the data itself. If the former aimed at properties of the model, the latter is looking at the properties of computational algorithms. SE practitioners need to be educated in data sciences to enable them to practice the above tools and methods as an integral part of SE.

Data analytics and IoT are wide-scope revolutions of digital surroundings. They create complex CPS that add new functionalities and capabilities to the existing physical environment. Designing an IoT system that has analytic capabilities involves “multi stack” layers, addressing SoS and network of networks.

SE practitioners should view a system as interconnected system elements performing the system functions. To meet this challenge, the SE practitioner who leads data-driven designs needs interdisciplinary knowledge of the

main aspects of IoT: computing, sensors/actuators, software, network, analytics and data science.

4.3.8 Service Systems

OASIS (2012) defines a service as:

A mechanism to enable access to one or more capabilities, where the access is provided using a prescribed interface and is exercised consistently with constraints and policies as specified by the service description.

It involves application of specialized competences (knowledge and skills) through deeds, processes, and performances for the benefit of another entity or the entity itself in real world. The entity involved with the service can be technical, socio-technical, or strictly social.

For service systems, understanding the integration needs among loosely coupled systems and system elements, along with the information flows required for both governance and operations, administration, maintenance, and provisioning of the service, presents major challenges in the definition, design, and implementation of services (Domingue, et al., 2009; Maier, 1998). Cloutier, et al. (2009) presented the importance of Network-Centric Systems (NCS) for dynamically binding different system entities in engineered systems rapidly to realize adaptive SoSs that, in the case of service systems, are capable of knowledge emergence and real-time behavior emergence for service discovery and delivery.

[Figure 4.9](#) illustrates the conceptual framework of a service system.

Typically, a service system is composed of service system entities that interact through processes defined by governance and management rules to create different types of outcomes in the context of stakeholders with the purpose of providing improved customer interaction and value cocreation.

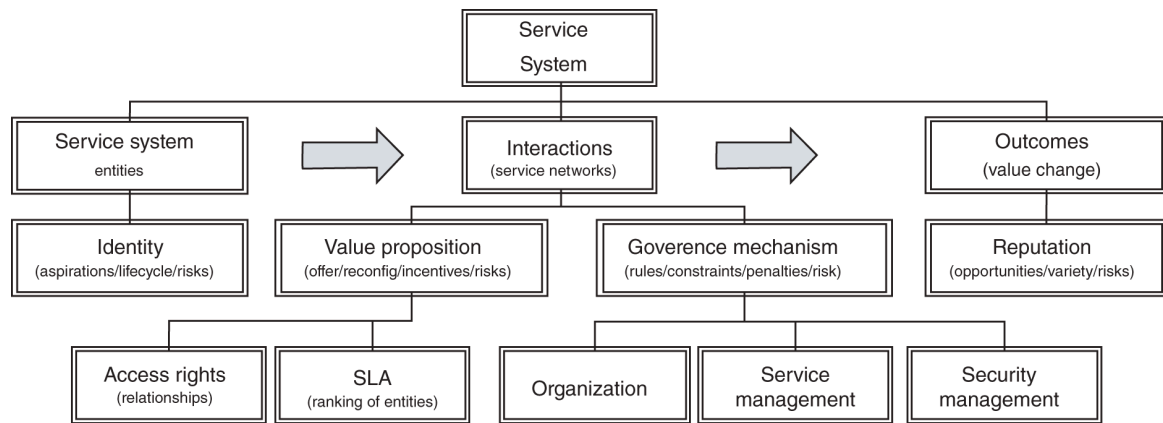


FIGURE 4.9 Service system conceptual framework. From Spohrer (2011). Used with permission. All other rights reserved.

Services not only involve the interaction between the service provider and the consumer to produce value, but have other intangible attributes like quality of service (e.g., ambulance service availability, response time to an emergency request). The demand for service may have loads dependent on time of day, day of week, season, or unexpected needs (e.g., natural disasters), and services are rendered at the time they are requested. Thus, the design and operations of service systems “is all about finding the appropriate balance between the resources devoted to the systems and the demands placed on the system, so that the quality of service to the customer is as good as possible” (Daskin, 2010).

In many cases, taking a service SE approach is imperative for the service-oriented, customer-centric, holistic view to select and combine service system entities to define and discover relationships among service system entities to plan, design, adapt, or self-adapt to cocreate value. Typically, five types of resources need to be considered: people; tangible products and environment infrastructure; organizations and institutions; protocols; and shared information and symbolic knowledge in the service delivery process. Major challenges include the dynamic nature of service systems evolving and adapting to constantly changing operations and/or business environments and the need to overcome silos of knowledge.

Interoperability of service system entities through interface agreements must be at the forefront of the service SE design process for the harmonization of operations, administration, maintenance, and provisioning procedures of the individual service system entities (Pineda,

2010). In addition, service systems require open collaboration among all stakeholders, but recent research on mental models of multidisciplinary teams shows integration and collaboration into cohesive teams has proven to be a major challenge (Carpenter, et al., 2010).

In summary, in a service system environment, SE practitioners should bring a customer focus to promote service excellence and to facilitate service innovation through the use of emerging technologies to propose creation of new service systems and value cocreation. SE practitioners must play the role of an integrator, considering the interface requirements for the interoperability of service system entities—not only for technical integration but also for the processes and organization required for optimal customer experience during service operations.

4.3.9 Enterprise Systems

This section illustrates the applications of SE principles and concepts when the SoI is an enterprise. The aim is to continuously improve and help transform the enterprise to better deliver value and to survive in a globally competitive environment. Enterprise SE is an emerging discipline that focuses on frameworks, tools, and problem-solving approaches for dealing with the inherent complexities of the enterprise including exploitation of new opportunities that can facilitate achievement of enterprise goals. A good overall description of enterprise SE is provided in Rebovich and White (2011). For more detailed information on this topic, please see the Enterprise SE articles in Part 4 of SEBoK (2023).

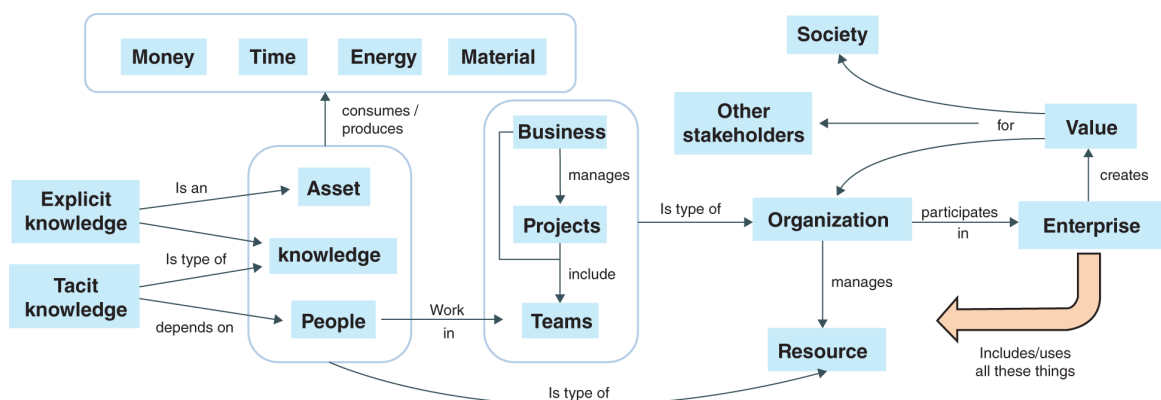
Enterprise

An enterprise consists of a purposeful combination (e.g., a network) of interdependent resources (e.g., people, processes, organizations, supporting technologies, and funding) that interact with each other to coordinate functions, share information, allocate funding, create workflows, and make decisions, and that interact with their environment(s) to achieve business

and operational goals through a complex web of interactions distributed across geography and time (Rebovich and White, 2011).

An enterprise must do two things: (1) develop things within the enterprise to serve as either external offerings or as internal mechanisms to enable achievement of enterprise operations, and (2) transform the enterprise itself so that it can more effectively and efficiently perform its operations and survive in its competitive and constrained environment.

It is worth noting that an enterprise is not equivalent to an “organization.” As shown in [Figure 4.10](#), an enterprise has organizations that participate in it, but these organizations are not necessarily “part” of the enterprise. The organizations that participate in the enterprise will manage a variety of resources for the benefit of the enterprise, such as people, knowledge, and other assets such as processes, principles, policies, practices, culture, doctrine, theories, beliefs, facilities, land, and intellectual property. These organizational resources will consume or produce money, time, energy, and material when acting on behalf of the enterprise.



Notes:

1. All entities shown are decomposable, except people. For example, a business can have sub-businesses, a project can have subprojects, a resource can have sub-resources, an enterprise can have sub-enterprises.
2. All entities have other names. For example, a program can be a project comprising several subprojects (often called merely projects). Business can be an agency, team can be group, value can be utility, etc.
3. There is no attempt to be prescriptive in the names chosen for this diagram. The main goal of this is to show how this chapter uses these terms and how they are related to each other in a conceptual manner.

FIGURE 4.10 Organizations manage resources to create enterprise value. From SEBoK (2023). Used with permission. All other rights reserved.

Creating Value

As shown in [Figure 4.10](#), an enterprise creates value for society, for other

stakeholders, and for the organizations that participate in that enterprise. It also shows other key elements that contribute to the value creation process. There are many types of organizations to implement value-creating enterprises: businesses (companies), networks of companies, programs and projects, virtual organizations, etc. A typical business may participate in multiple enterprises through its portfolio of projects. A large SE project can be an enterprise in its own right (implemented as a virtual organization), with participation by many different businesses, and may be organized as a number of interrelated subprojects. In many cases, enterprises find themselves in a rapidly changing environment where stakeholder needs change over time. Therefore, an enterprise must constantly adapt its capabilities to meet the enterprise strategic goals and objectives.

Capabilities in the Enterprise

As shown in [Figure 4.11](#), the enterprise acquires or develops systems or individual elements of a system. The enterprise can also create, supply, use, and operate systems or system elements. Since there could possibly be several organizations involved in this enterprise venture, each organization could be responsible for particular systems or perhaps for certain kinds of elements. Each organization brings their own organizational capability with them, and the unique combination of these organizations leads to the overall operational capability of the whole enterprise.

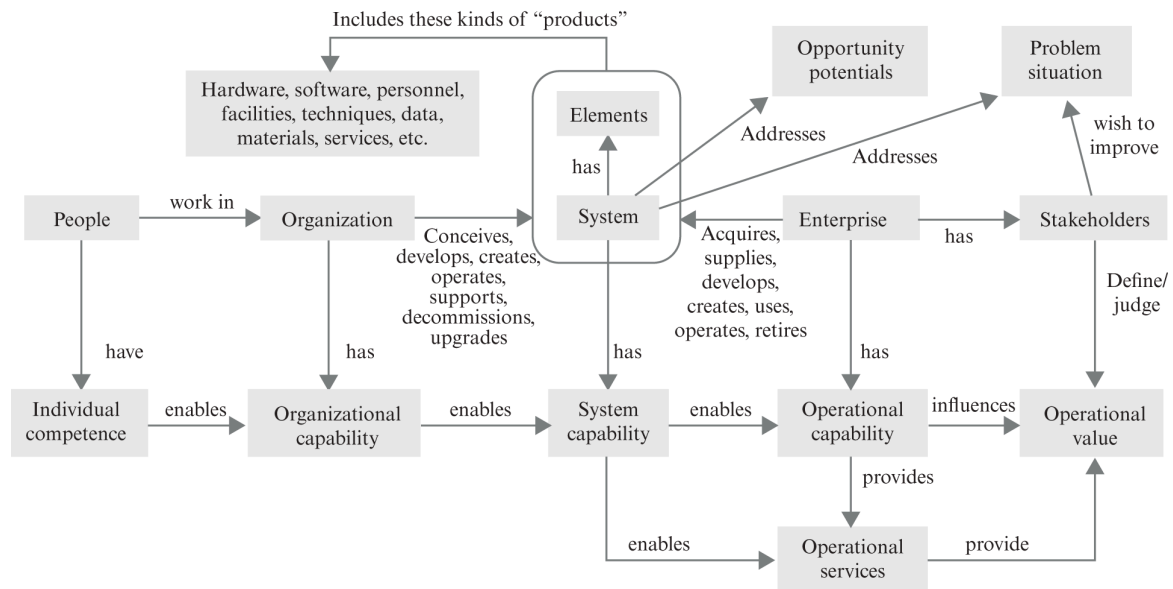


FIGURE 4.11 Individual competence leads to organizational, system, and operational capability. From SEBoK (2023). Used with permission. All other rights reserved.

The word “capability” is used in SE in the sense of “the ability to do something useful under a particular set of conditions.” This section discusses three different kinds of capabilities: organizational capability, system capability, and operational capability. It uses the word “competence” to refer to the ability of people relative to the SE task. Individual competence (sometimes called “competency”) contributes to, but is not the sole determinant of, organizational capability. This competence is translated to organizational capabilities through the work practices that are adopted by the organizations. New systems (with new or enhanced system capabilities) are developed to enhance enterprise operational capability in response to stakeholder’s concerns about a problem situation.

As also shown in [Figure 4.11](#), operational capabilities provide operational services that are enabled by system capabilities. These system capabilities are inherent in the system that is conceived, developed, created, and/or operated by an enterprise. Enterprise SE concentrates its efforts on maximizing operational value for various stakeholders, some of whom may be interested in the improvement of some problem situation.

Enterprise SE, however, addresses more than just solving problems; it also deals with the exploitation of opportunities for better ways to achieve the enterprise goals. These opportunities might involve lowering operating

costs, increasing market share, decreasing deployment risk, reducing time to market, and any number of other enterprise goals. The importance of addressing opportunity potentials should not be underestimated in the execution of enterprise SE practices.

The operational capabilities of an enterprise will have a contribution to operational value (as perceived by the stakeholders). Notice that the organization or enterprise can deal with either the system as a whole or with only one (or a few) of its elements. These elements are not necessarily hard items, like hardware and software, but can also include “soft” items, like people, processes, principles, policies, practices, organizations, doctrines, theories, beliefs, and so on.

Enterprise Drivers and Outcomes

An enterprise needs to consider its own needs that relate to enabling assets (e.g., personnel, facilities, communication networks, computing facilities, policies and practices, tools and methods, funding and partnerships, equipment and supplies) when addressing the stakeholders’ needs. The purpose of the enterprise’s enabling assets is to effect state changes to relevant elements of the enterprise necessary to achieve targeted levels of performance. The enterprise “state” shown in [Figure 4.12](#) is a complex web of past, current, and future states (Rouse, 2009). The enterprise work processes use these enabling assets to accomplish their work objectives to achieve the desired future states.

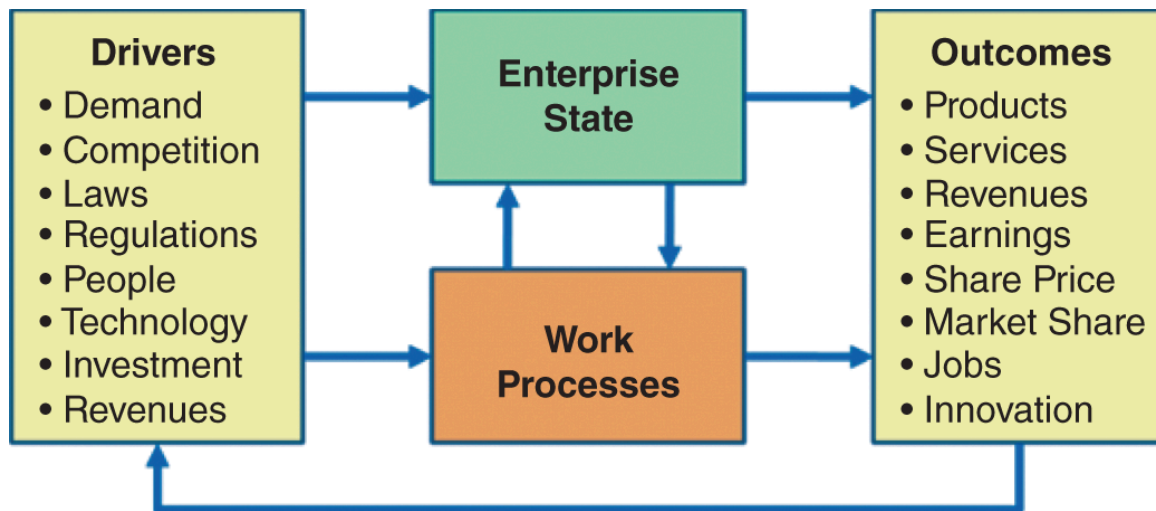


FIGURE 4.12 Enterprise state changes through work process activities. From rouse (2009). Used with permission. All other rights reserved.

Since a high degree of complexity is to be assumed, it is advisable to apply formalized modeling methods to achieve the enterprise strategic goals and objectives. It has proven useful to use enterprise architecture analysis to model these states and the relative impact each enabling asset has on the desired state changes. This analysis can be used to determine how best to fill capability gaps and minimize the excess capabilities (or “capacities”). The needs and capacities are used to determine where in the architecture elements need to be added, dropped, or changed. Each modification represents a potential benefit to various stakeholders, along with associated costs and risks for introducing that modification.

Enterprise Opportunities and Opportunity Assessments

The potential modifications that are identified represent opportunities for improvement. Usually, these opportunities require the investment of time, money, facilities, personnel, and so on. There might also be opportunities for “divestment,” which could involve selling of assets, reducing capacity, canceling projects, and so on. Each opportunity can be assessed on its own merits, but usually these opportunities have dependencies and interfaces with other opportunities, with the current activities and operations of the enterprise, and with the enterprise’s partners. Therefore, the opportunities may need to be assessed as a “portfolio,” or, at least, as sets of related opportunities. Typically, a business case assessment is required for each

opportunity or set of opportunities. If the set of opportunities is large or has complicated relationships, it may be necessary to employ portfolio management techniques. The portfolio elements could be bids, projects, products, services, technologies, intellectual property, etc., or any combination of these items. Examples of an enterprise portfolio captured in an architecture modeling tool can be found in Martin (2005), Martin et al. (2004), and Martin (2003).

The results of the opportunity assessment can be compiled and laid out in an enterprise plan (sometimes conveyed as an enterprise ConOps) that considers all relevant factors, including system capabilities, organizational capabilities, funding constraints, legal commitments and obligations, partner arrangements, intellectual property ownership, personnel development and retention, and so on. The plan usually goes out to some long horizon, typically more than a decade, depending on the nature of the enterprise's business environment, technology volatility, market intensity, and so on. The enterprise plan needs to be in alignment with the enterprise's strategic goals and objectives and leadership priorities.

Practical Considerations

When it comes to performing SE at the enterprise level, Rebovich and White (2011) provide several good practices:

- Set enterprise fitness as the key measure of system success. Leverage game theory and ecology, along with the practices of satisfying and governing the commons.
- Deal with uncertainty and conflict in the enterprise through adaptation: variety, selection, exploration, and experimentation.
- Leverage the practice of layered architectures with loose couplers and the theory of order and chaos in networks.

Enterprise governance involves shaping the political, operational, economic, and technical landscape. One should not try to control the enterprise like one would in a traditional SE effort at the project level.

4.4 APPLICATION OF SYSTEMS ENGINEERING FOR SPECIFIC PRODUCT SECTOR OR DOMAIN APPLICATION

This chapter presents how SE is applied in different product sectors or application domains. For each of these, unique and domain-specific terms, concepts, activities, methods, and practices are introduced.

The following domains are presented in alphabetical order:

- Automotive Systems;
- Biomedical and Healthcare Systems;
- Commercial Aerospace Systems;
- Defense Systems;
- Infrastructure Systems;
- Oil & Gas Systems;
- Power & Energy Systems;
- Space Systems;
- Telecommunication Systems;
- Transportation Systems.

Note that the application of SE is not limited to the product sectors and application domains listed above. SE is a generic discipline that can be applied in most situations and domains (with varying levels of value). However, the details of how the practice is applied will vary in different product sectors or application domains.

4.4.1 Automotive Systems

Overview of SE Applications within the Automotive Domain

The automotive industry has a long history of engineering complicated and more and more complex consumer products, characterized by diverse product ranges, high production volumes, and a very competitive market. To make vehicles attractive, manufacturers have to balance efficiency for

mass production with driving performance. Strong, well-orchestrated processes across the supply chain are key to meeting this challenge. Quality is highly dependent on good processes being followed rigorously, and economic performance relies on optimization. Electrification, connectivity, autonomy, and consumer choice are driving complexity, creating opportunities for SE to enter the mainstream in coming years. The INCOSE Automotive SE Vision 2025 document (2020) provides an excellent summary of the current and future trends in this domain. See [Table 4.5](#) for a comparison of automotive with two other domains considering some of the characteristics that may affect SE approaches.

TABLE 4.5 Comparison of automotive, aerospace/defense, and consumer electronics domains

	Automotive	Aerospace/ Defense	Consumer Electronics
Customer requirements	Assumed by manufacturer	Defined by customer	Assumed by manufacturer
Legislative environment	Certification/inspection of product design, auditing of development and production processes (e.g., ISO/TS 16949 (2009)/IATF 16949, ISO 26262 (2018), E-marking, conformity of production, many regulatory Inc) standards (UNR, FMRSS, etc.) applicable to the overall system and its elements. Process regulations/standards are a new development.	Certification (e.g., DO-178C from RTCA Inc) in process, tooling and product. Auditing of development and production processes	CE/UL mark according to a standard (typical per product)
User skill level	Somewhat trained	Highly trained	Untrained
Complexity (PLE/component reuse)	High	Low	Medium

	Automotive	Aerospace/ Defense	Consumer Electronics
Complexity (sociodynamic)	High	High	Low
Product development cycle time	Medium (3–7 years)	Long (10+ years)	Short (years)
Delivery cadence	Annual	Decade	Annual
Product design life	Medium (10–20 years)	Long (30+ years)	Short (years)
Approach to maintenance	Repair	Repair	Replace
Connectivity need	Medium (trend to high)	Low-medium (defined in stakeholder requirements)	Medium
Number of external integration interfaces	Medium (trend to high)	Medium (trend to high)	Medium
Safety/cybersecurity criticality	High	High	Medium
Typical industry operating margin	4–6%	6–8%	8–10%

	Automotive	Aerospace/ Defense	Cons Elect
Annual production volume	10k–900k	100–100k	100k

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Emergence of SE in the Automotive Domain

Since General Motors vehicles first shared a common chassis in 1908, the automotive industry has employed extensive reuse in combination with variant management techniques to manage costs and keep delivery cycles short. Until the late 1960s, activities familiar to the SE domain centered on parts development, promoting reuse through standards and matrix organizations. Architects mainly addressed geometry until the on-board electronics and software revolution started in the 1970s. Standalone mechatronic and automated control systems appeared first, delivering applications such as engine control, anti-lock braking, and automatic heating and air conditioning. During the 1990s, vehicles acquired extensive networks of interconnected electronic control units. In the same period, many manufacturers invested heavily in engineering teams that focused on elicitation and refinement of stakeholder needs to address emergent properties such as safety, environmental impact, dynamic performance, and occupant comfort. This subset of activities (e.g., stakeholder requirements, electrical and electronic architecture) provided the ingredients for the emergence of SE in automotive.

Contemporary SE in the Automotive Domain

Increasing complexity, driven by the parallel trends of electrification, on- and off-board networks, automation, and autonomy has led to growing interest in automotive SE since 2000. Electronics and software began to

outstrip mechanical design as a means for manufacturers to provide distinctive products. As shown in [Table 4.6](#), the automotive industry is very standards-driven and the last decade has seen many new developments. These standards address, in particular, architectural approaches, software elements, safety, and security processes. Practices have evolved in isolated pockets, leading to significant variation in how the discipline is interpreted and executed. Convergence of capabilities at different maturity levels, led by engineering executives with different worldviews and backgrounds, has led to the adoption of many different SE paradigms with varying enthusiasm. SE practitioners might draw the system boundary at the vehicle level, or be restricted to applying their techniques to individual features, system elements, or domains, sometimes focusing on electronics and software. Approaches differ too, with some being requirements-led, others architecture-centric. SE has entered the automotive domain in incremental steps, and continues to do so, due to the need to accommodate the special features and legacy of each company. Disruptive change is hard for established players because their business models have low margins, and rely on trusted, repeatable processes that leverage extensive reuse of system elements. Similarly, production needs efficient processes to continuously deliver high volumes of system elements and assemblies on a just-in-time basis, making interruptions to manufacturing difficult to tolerate.

TABLE 4.6 Representative organizations and standards in the automotive industry

Organization/standard	Description
SAE International, formerly the Society of Automotive Engineers	One of the main organizations that coordinate the development of technical standards for the automotive industry. Currently, SAE International is a globally active professional association and standards organization for engineering professionals in various industries, whose principal emphasis is placed on transport industries such as automotive, aerospace, and commercial vehicles
Japan Society of Automotive Engineers (JSAE)	An organization that sets automotive standards in Japan, analogous to the SAE
Association for Standardization of Automation and Measuring	An incorporated association under German law whose members are primarily international car manufacturers, suppliers, and engineering service providers from the automotive industry. The ASAM standards define protocols, data models, file formats, and application programming interfaces (APIs) for the use in the development and testing of automotive electronic control units
AUTomotive Open System ARchitecture (AUTOSAR)	An open and standardized automotive software architecture, jointly developed by automobile manufacturers, suppliers, and tool developers. Some of its key goals

Organization/standard	Description
	include the standardization of basic system functions, scalability to different vehicle and platform variants, transferability throughout the network, integration from multiple suppliers, maintainability
The GENIVI Alliance	A nonprofit consortium whose goal is to establish a globally competitive, Linux-based operating system, middleware, and platform for the automotive in-vehicle infotainment (IVI) industry. GENIVI specifications cover the entire product life cycle and software updates and upgrades over the vehicle's lifetime
ISO/TS 16949 (2009)/IATF 16949	An international standard for particular requirements for the application of ISO 9001 quality management systems for automotive production and relevant service part organizations
IEC 62196 (2022)	An international standard for set of electrical connectors and charging modes for electric vehicles maintained by the International Electrotechnical Commission (IEC)
ISO 26262 (2018)	Road vehicles - Functional safety
ISO/SAE 21434 (2021)	Road vehicles - Cybersecurity engineering

Organization/standard**Description**

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The explosion of electrical, electronic, and software systems is driving awareness that engineering based on assembling parts from suppliers without a systems approach is no longer enough. Full autonomy is on the horizon, and advanced driver assistance features like lane keeping and emergency braking are well established. Many new vehicles have high-integrity system elements linked to the outside world. Managing the complexity and cybersecurity risks this creates is a great challenge (see [Section 6.5](#)). Cultural and methodological changes are ongoing in industry incumbents. They increasingly face competition from newcomers with backgrounds in software-intensive industries (see [Section 4.3.4](#)), who in turn must adapt their culture and scale their business models to the realities of high-volume automotive manufacturing.

New Eco-Systems Involving the Automotive Industry: The Example of “Mobility as a Service”

In many urban areas, local governments implement policies to reduce the number of private cars and to foster the deployment of mobility services as a complement to public transport. The number of privately owned, individual cars has decreased dramatically in some big cities. For instance, in Paris, capitol of France, the percentage of cars as a mode of transportation has decreased from 46% in 2002, to only 13% in 2022. Vehicle manufacturers should not expect that this trend will change. Vehicles they used to sell are more and more being replaced by public transport, biking, and walking, as well as services still involving vehicles like car-sharing or ridesharing. All these service offers can be integrated into a larger service enabling them to combine and thus making on-demand mobility faster and easier. This is called Mobility as a Service

(MaaS), with a lot of initiatives around the world triggered by the Sustainable Development Goals introduced in [Section 3.1.10](#). However, MaaS is not a silver bullet nor standard yet. MaaS may be considered as the mission for an SoS involving mobility service operators, both public and private. They have to cooperate in order to offer a more attractive user experience and at the same time to make the conditions for local mobility more sustainable.

The Future of SE in the Automotive Domain

Dealing with massively expanding complexity in an extremely challenging environment where standards and regulation trail fast-paced innovation is a challenge for this process-driven industry, but an opportunity for SE. As connectivity and autonomy become the norm, vehicles are built on highly configurable software platforms for providing mobility as a service. Development cycles that took five years are being compressed, where service updates that took a year will be expected in weeks. Delivering change like this means fundamental shifts in thinking are required across the board: from new business models, through service-centric architectures, to security-informed safety paradigms. SE is the means by which this can be achieved.

4.4.2 Biomedical and Healthcare Systems

Overview of SE Applications within the Biomedical and Healthcare Domain

SE has become more important to the healthcare industry (SEBoK, 2023), especially as systems and processes get more complex and quality characteristics such as safety, security, reliability, and human systems integration become more challenging. SE offers numerous benefits to biomedical and healthcare systems including the following:

- Supports design and development of healthcare systems using well-defined processes and standards,

- Offers well-defined approaches to design and implement architectures for proper interfacing, networking and communications using open industry standards,
- Enables operators and enterprises to scale up without compromising quality of operations,
- Enables better insights and control of many production systems including quality assurance, inventory, and cost control, and
- Augments user experience of various stakeholders like doctors, and surgeons by system level integration of emerging digital platforms like augmented reality, virtual reality, and robotics.

In the medical industry, especially for medical devices, it is important to understand that “risk management” is generally centered around product (user safety) risk and (called system safety in this handbook—see [Section 3.1.11](#)) rather than project (technical or business) risk (called risk management in this handbook—see [Section 2.3.4.4](#)).

Unique Considerations for Healthcare Delivery

SE applied to healthcare delivery differs significantly from conventional SE as applied in traditional fields such as defense, aerospace, and automotive. Most healthcare delivery projects involve improvement of an imperfect workflow or care process or the design of a limited scope new workflow or care process in a local clinic, hospital, laboratory, or in population health. If successful, solutions are shared with peer institutions in the same medical organization. As a result, most SE projects in healthcare delivery involve only a few stakeholders and a handful of requirements. Approaches leveraging lean SE have shown to be successful in many cases (Oppenheim, 2021) (see [Section 4.2.3](#)). Healthcare delivery operations have a critical need for the SE process to address pervasive healthcare problems such as care fragmentation (e.g., the systemic misalignment of incentives) or lack of coordination that spawn inefficient allocation of resources or harm to patients. Just as in medical device development, SE in healthcare delivery also strongly emphasizes patient safety. Methods such as the Systems Engineering Intervention for Patient Safety (SEIPS) (Carayon, 2006) focus

on tailoring SE processes to the specific context of patient-centered medicine.

Unique Considerations for Medical Devices

In contrast to healthcare delivery systems, some medical device and healthcare IT companies use a more traditional form of SE. However, some are heavily tailoring SE approaches to incrementally demonstrate the value of SE. Many devices must work in harsh environments, including inside the human body. Interoperability, interconnectivity, and transportability are increasingly critical for medical devices and SaMDs (Software as Medical Devices). During audits and submissions, regulators require device developers to follow standard quality system processes (e.g., ISO 13485). Standards such as ISO 14971 (application of risk management to medical devices), IEC 60601 (medical device safety), IEC 62304 (Medical Device software—Software life cycle processes), and IEC 62366 (application of usability engineering to medical devices) are driving medical device organizations to take a deeper look into system safety and the engineering practices behind it. Thus, SE practitioners are increasingly being brought on board to leverage their life cycle management skills and support validating that the final product does indeed meet the needs of its stakeholders. In addition to an emphasis on systems safety, the medical device sector is seeing an increasing need for several SE methodologies including, but not limited to, SoS management, stakeholder management, agile systems development, trade analysis, MBSE, and PLE.

Unique Activities, Methods, and Practices

Healthcare Systems are often broad in context including a population of diverse patients, many healthcare professionals, many medical devices, many insurance companies, many delivery systems, regulators, and the government. One emphasis of SE practitioners in the biomedical and healthcare domain is patient safety risk, often more so than technical or business risks (see [Section 6.1](#)). Traceability is often a key factor in regulatory submissions and audits. Organizations that have strong SE

practices are therefore in a better position to avoid pitfalls and to effectively defend their decisions if a regulatory audit does occur. In general, applicable standards do not need to be excessively tailored, although organizations with new or maturing practices may want to focus on lean implementations to obtain early and effective system adoption. Carefully balancing the trades between healthcare costs, better health outcomes for populations, and profits for shareholders is an ongoing challenge for Healthcare SE practitioners. On a larger scale, healthcare SE practitioners that can influence policy and incentives will become even more valuable to their organizations.

4.4.3 Commercial Aerospace Systems

Overview of SE Applications within the Commercial Aerospace Domain

SE is part of the strategies for the development of solutions and products in the commercial aerospace system domain. Commercial aerospace systems are complex, and their complexity continues to increase. The increased use of software makes it possible to implement more functions than before, which contributes to a further increase in complexity. At the same time, the expectation is raised that the increased use of software will make solutions and products available more quickly than the historic mechanical systems of the past. As shown in [Figure 4.8](#), commercial aerospace systems are often part of larger SoSs. Future commercial aerospace systems will include autonomy, artificial intelligence, neural networks, novel propulsion, advanced human system integration (HSI), and cybersecurity.

Commercial aerospace systems use sequential as well as incremental and evolutionary life cycle models, including agile methods with smaller cycles. Thus, the processes in this handbook can be used to address and help organizations manage these new factors derived from complexity settings. The adoption of new technologies and perspectives emphasize some concepts such as the systemic approaches and use of SoS approaches to support organizations by putting them on the forefront of the market with

competitive products, adapted to the new reality of increasing interoperability.

Unique Terms & Concepts

The commercial aerospace organizations of many countries have specific policies, standards, and guidebooks to guide the application of SE in their organizational environment. For example, ARP 4754A (2010) describes the standard practices for verifying commercial aircraft requirements.

There are many other systems related to this domain. For example, in the aviation domain, there are systems that go far beyond the aircraft itself according to the interaction characteristic of system elements described on system concept definition. Examples include an air traffic control system.

New applications of commercial aerospace systems are being continually introduced. For example, some organizations specifically created to address the new aerospace segment of flying cars have started a great race in a totally different way. By using new methodologies and approaches, these systems are being developed by considering their integration in completely new context. The same is happening with unmanned and autonomous vehicles. These new applications require an understanding of the ecosystem of the new operational contexts, as well as the lifestyles of its users.

Unique Activities, Methods, and Practices

SE may help the realization of effective commercial aerospace systems through the following activities, methods, and practices:

- **Stakeholders.** Stakeholders vary greatly and can range from federal government services, to aircraft manufacturers, to passengers.
- **Design and construction practices.** Model-based design is generally used from construction model specifications, which enables and maintains traceability between requirements and models.

- **Interfaces.** Because commercial aerospace systems' system elements are developed in various parts of the world and brought to a single (or multiple) location for assembly, adherence to interface management principles is critical.
- **Risk management.** Risk management is essential, especially for the introduction of new technologies.
- **Safety.** Finally, it is important for SE management to assure that safety is not compromised by organizational factors, as described by Paté-Cornell (1990).

Examples of how SE helps in resolving unique domain challenges include:

For aircraft original equipment manufacturers (OEMs), SE:

- Helps in design and manufacturing of aircraft subsystems, assembly and integration testing using well-defined process, standards, and quality standards.
- Offers well-defined approaches to create designs or architectures, processes, and roadmaps for proper interfaces, instrumentation, and communications that enable better visibility of the static and dynamic operational data and status of the subsystems.
- Enables operators and enterprises to scale up without compromising quality of production using a well-defined SE framework, tools, and emerging technologies.
- Enables better insights and control of congestion and traffic control of many schedules like flights, passenger, luggage, and food.
- Augments user experience of various stakeholders by system level integration of emerging digital technologies like augmented reality and virtual reality for enriched cockpit and instruments.

For airlines, SE:

- Helps in the support stage, to maintain the fleet.
- Helps balance performance and environmental impacts.

- Offers a set of procedures and activities to manage the services that consider human resources, information, and operation data.

Other Unique Considerations

SE is increasingly being applied in commercial practice. Petersen and Sutcliffe (1992), for example, discuss the principles of SE as applied to aircraft development. Life cycle functions of the commercial aerospace industry gives SE its own unique characteristics.

4.4.4 Defense Systems

Overview of SE Applications within the Defense Domain

While SE has been practiced in some form from antiquity, what has now become known as the modern definition of SE has its roots in defense systems of the twentieth century. It became recognized as a distinct activity in the late 1950s and early 1960s due to technological advances taking place that led to increasing levels of system complexity and systems integration challenges, and the need for SE further increased with the large-scale introduction of digital computers and software.

SE within defense evolved to address systemic approaches to issues such as the widespread adaptation of COTS technologies and the use of SoS approaches. It offers well-defined designs/architecture, processes and roadmaps for proper interfacing, networking, and communications. This enables better integrity and interoperability of real-time intelligence data across various devices, from various vendors, and platforms using open industry standards. Today, with increasing emphasis on networks and capabilities the defense organizations of many countries are recognizing the criticality of end-to-end SoS performance and increasing focus on integration to deliver these capabilities.

Unique Considerations

Defense systems have numerous characteristics and consequently, a huge complexity, making SE essential for their development:

- They are complex technical systems with many stakeholders and compressed development timelines.
- The systems must be highly available and work in extreme conditions all over the world—from deserts to rain forests and to arctic outposts.
- There are long system life cycles, so logistics is of prime importance.
- There is typically a strong human interaction, so usability/human systems integration is critical for successful operations.
- There is at times a need for defense operators and enterprises to accelerate development and production (e.g., quick response in event of national emergency or increased threats) without compromising quality of operations using a well-defined SE framework, tools, and emerging technologies

Unique Activities, Methods, and Practices

SE has a strong heritage in defense, and much of the SE processes in this handbook can be used as is in a straightforward manner, with normal project tailoring to address unique aspects of the project. It is important to note that as ISO/IEC/IEEE 15288 (2023) has evolved into a more domain- and country-neutral SE standard, so care must be taken to ensure that the defense focus is reasserted upon application. An example of specific implementation of ISO/IEC/IEEE 15288 when utilized for US Department of Defense projects is provided in IEEE 15288.1 (2014). This standard provides the basis for selection, negotiation, agreement, and performance of necessary SE activities and delivery of products. Additionally, the standard allows flexibility for both innovative implementation and tailoring of the specific SE processes to be used by system suppliers, either contractors or government system developers, integrators, maintainers, or sustainers. The defense organizations of many countries also have specific policies,

standards, and guidebooks to guide the application of SE in their environment.

4.4.5 Infrastructure Systems

Overview of SE Applications within the Infrastructure Domain

This section addresses physical capital projects infrastructure including public works, transport, complex buildings, and industrial facilities. Within the infrastructure domain, SE practices are more developed in the high-technology system elements that involve software development, control systems, system security, or system safety. Infrastructure projects tend to define the high-level design solution without requirements decomposition, allocation, or interface identification. Architectures, traceability, and relationships within the project are often implied rather than specified. Infrastructure owners can benefit by applying SE to provide systematic, formal, verifiable connections between the business needs and the final product.

Unique Terms and Concepts

Infrastructure projects are distinguished from manufacturing and production, as they usually focus on unique, large physical systems where construction takes place on site rather than in a factory. These projects are adapted and integrated to existing environments, and are often characterized by loosely defined boundaries, evolving system architectures, multiphase implementation efforts which can exceed a few decades, and multiple-decade asset life cycles. As a result, stakeholders' expectations and design solutions evolve over an extended timeframe. Unlike other SE domains, most infrastructure projects cannot be standardized and do not involve a prototype.

Many of the processes described in this handbook can be used to manage infrastructure projects but in some cases with different terminology, as

illustrated in [Table 4.7](#). There are some areas where existing infrastructure practices could be adjusted slightly to better align with SE practices.

TABLE 4.7 Infrastructure and SE definition correlation

Systems Engineering Term	Infrastructure Term	Recommendation
Acquirer	Owner or Agency	
Acquisition	Contracting phase; Procurement	Share good practices and lessons learned to improve procurement documents to enable better owner control of the project.
Business requirements	Project need; Business case	Derives contractor requirements from the business requirements, hold requirement reviews and include in the contractors' scope.
Configuration control	Versioning	Configuration identification, change management, status accounting, configuration audit according to ISO 10007 (2017).
Decision gate	Milestone	Clearly define entry and exit criteria for decision gates
Life cycle	Project life cycle	Include how the infrastructure will deliver its intended function and long-term asset management. Add in contractor's scope expectations

Systems Engineering Term	Infrastructure Term	Recommendation
		that will benefit the entire project life cycle.
Performance requirement	Often found in Technical Specifications	Allocate top-level system performance requirements to system elements, defining performance requirements. Best performed by the acquiring entity unless the procurement method is a PPP.
Requirements	Design Criteria; Scope of Work; or Specifications	Integrate full life cycle considerations into design criteria, including operations, maintenance, and disposal/replacement planning.
Supplier	Design Consultant; Contractor	Use requirements management to strengthen procurement language and enforce contract requirements during the project. Clearly define acceptance criteria and performance measures.

Systems Engineering Term	Infrastructure Term	Recommendation
System architecture	Context diagram; Schematics; Process and Instrumentation Diagrams	Consider creating early in project life cycle to support requirement allocation and interface management. Use ICDs or N ² diagrams to complement the system architecture.
Verification	Design Review; Quality Control (QC)/Quality Assurance (QA)	Provide sufficient schedule and budget for both QC and Q A activities, including specific audit periods and “pens down” dates for each milestone. In the design phase, confirm the design meets requirements (QC), and procedures were followed (QA)
Validation	Construction Inspection; Quality Control (QC) / Quality Assurance (QA)	Include sufficient budget and authority for QA to ensure compliance. Ensure acceptance testing refers back to stakeholder needs and includes a focus on whether it meets its intended use.

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Unique Activities, Methods, and Practices

SE may help the realization of effective infrastructure systems through the following activities, methods, and practices:

- **Stakeholders.** Stakeholders can range from governmental legislators who control funding for the project, to local/regional agencies that add beautification needs, to landowners with adjacent property impacted by a proposed project. In all government-funded projects, segments of the public may also be a stakeholder group. The wide array of potential stakeholders makes requirements gathering, cost, and schedules volatile. Public and political pressure can cause premature initiation of projects, with incomplete project scope and ill-defined metrics.
- **Design and construction practices.** Within infrastructure, the engineering disciplines have well-established, traditional practices and are guided by independent industry codes and standards that are not shared between disciplines. Design requirements are generally dissociated from construction specifications, therefore limiting traceability between design and construction.
- **Interfaces.** Infrastructure projects have external, often uncontrollable interfaces that can impact the project. Interfaces can include existing built systems, natural systems, environmental, and other internal and external dynamics.
- **Risks.** The contractual framework and allocation of liability and commercial risk are major factors impacting procurement and contracting processes. SE practices may therefore help to manage risk associated with cost estimating, changing scope, system integration, and verification. They may also improve construction productivity, making infrastructure development more cost-effective.

Other Unique Considerations

SE concepts are relatively newly applied in the infrastructure domain. As the application of SE grows within the infrastructure domain, an effort should be made to train engineering discipline specialists in SE concepts.